Actual harmonic forms force whole-filling Milnor matching

24 September 2026. This note strengthens the realization criterion in Orbifold smoothing bridge, Theorem C. It replaces its assumed relative diffeomorphism of singular fillings by a condition on the **actual harmonic representative and the topology of the whole filling**. It does not assume that a terminal ALE bubble is the whole filling, or require a single-scale degeneration.

The new connection in this project is analytic-to-symplectic: actual-form alignment gives canonical contact boundaries; a zero-free harmonic form makes the entire filling symplectic; classical filling theorems then force its relative smooth type. The filling classifications are not new results of this project. Global Q-Gorenstein smoothability remains a separate input.

1. The input and the forced-matching theorem

Let (M_j, g_j, a_j) satisfy the hypotheses of Theorem A of the orbifold smoothing bridge. Explicitly, the M_j are closed connected oriented smooth four-manifolds, each a_j is an actual conformally solvable Spin^c class, and, for fixed $D, A_0 > 0$,

$$\operatorname{diam}(M_j, g_j) \leqslant D, \qquad p_j = (a_j^+)^2 \geqslant A_0, \qquad \delta_j + q_j \longrightarrow 0, \tag{1}$$

where

$$\delta_j = \| \operatorname{Ric}_{g_j}^\sharp + I \|_{\infty,\mathrm{op}}, \quad q_j = \frac{\mathfrak{D}_2(g_j, a_j)}{V_j}, \quad V_j = \operatorname{Vol}(M_j, g_j),$$

$$\mathfrak{D}_2(g, a) = \frac{3}{2} \int_M (s^2 + 12|W^+|^2)\, d\mu - 72\pi^2 (a^+)^2.$$

The structure determining a_j is fixed while auxiliary positive weights vary. No hypothesis on π_1 or π_2 is imposed.

Pass to the subsequence supplied by that theorem. Its limit is a compact negative Kähler–Einstein orbifold (X, h, J, ω_∞), with finitely many isolated quotient singularities x_ν. Choose small disjoint uniformizing neighborhoods U_ν, with spherical links Y_ν, and retain a collar on each side of Y_ν in the regular locus. The regular-locus embeddings identify the exterior

$$K = X \setminus \bigcup_\nu \operatorname{int} U_\nu$$

with a submanifold of M_j. Denote its complementary compact pieces by $B_{\nu,j}$; each has the actual boundary marking by Y_ν. These are the **whole singular fillings**. Any smaller bubbles or necks inside them are included.

Let H_j be the harmonic self-dual representative of a_j^+, not the harmonic representative of the whole class a_j, and put

$$\Omega_j = -2\pi H_j. \tag{2}$$

Theorem 1 (harmonic/contact forced matching). In the setting (1), suppose that, for all sufficiently large j,

1. H_j has no zero on any $B_{\nu,j}$;
2. for each ν, either
   - x_ν is an A_1 singularity and $b_2(B_{\nu,j}; \mathbb{Q}) = 1$, or
   - $B_{\nu,j}$ is a rational homology four-ball.

Then every $B_{\nu,j}$, with its actual boundary marking, is oriented diffeomorphic relative to that marking to a local Q-Gorenstein Milnor fiber of (X, x_ν). In the second alternative the singularity is **necessarily a Wahl singularity**

$$\frac{1}{n_\nu^2}(1, n_\nu a_\nu - 1), \qquad n_\nu \geqslant 2, \quad 0 < a_\nu < n_\nu, \quad \gcd(n_\nu, a_\nu) = 1. \tag{3}$$

In particular, cyclicity and local Q-Gorenstein smoothability need not be separately assumed in the rational-ball alternative.

The zero-free condition is only qualitative. There is no required uniform lower bound for $|H_j|$ on the shrinking pieces. It is a condition on a specified solution of a linear global elliptic equation, not a proposed relative diffeomorphism and not a choice of a convenient ALE model.

The proof occupies Sections 2–5. The analytic hypotheses alone do not presently prove condition 1. In contrast, nonvanishing and the canonical contact structure on the **fixed regular collars** will be consequences of (1), rather than additional assumptions.

2. Actual harmonic alignment upgrades on every regular collar

Lemma 2 (strong convergence of the actual harmonic forms). Using the same regular-locus embeddings as for the metrics,

$$\Omega_j \longrightarrow \omega_\infty \quad \text{in } C^{1,\beta}_{\mathrm{loc}}(X_{\mathrm{reg}}) \quad \text{for every } \beta < 1. \tag{4}$$

This needs neither the core zero-free assumption nor a negative-square budget.

Proof. Set

$$r_j = \fint |\mathrm{Ric}_{g_j} + g_j|^2, \qquad \kappa_j = r_j + q_j \longrightarrow 0.$$

The inherited actual-form composition, Theorem A, supplies smooth self-dual forms ω_j, with $\|\omega_j\|_\infty \leqslant 2\sqrt{3}$, such that, for $\eta_j \downarrow 0$,

$$\fint |\omega_j - \Omega_j|^2 \leqslant 49\kappa_j + \eta_j. \tag{5}$$

It also supplies the amplitude and first-derivative estimates used in the orbifold compactification proof. Since $V_j \leqslant V_0(D)$, (5) is strong unnormalized L^2 alignment. That proof constructs $\omega_j \to \omega_\infty$ strongly in local L^2. Thus the same local L^2 convergence holds for Ω_j, with uniform local L^2 bounds.

On nested regular harmonic-coordinate charts, the metrics converge in $C^{1,\alpha}$ and strongly in $W^{2,p}$ for every finite p. The forms Ω_j solve the actual homogeneous Hodge equation

$$(d + d^*_{g_j})\Omega_j = 0. \tag{6}$$

Interior first-order elliptic estimates for (6), initially with exponent 2, give a uniform local $W^{1,2}$ bound. Sobolev embedding gives a local L^4 bound. Iterating the same interior estimate gives $W^{1,4}$, hence L^p for every finite p, and then $W^{1,p}$ for these p. Constants are uniform on the chosen charts: their leading coefficients are uniformly $C^{1,\alpha}$, their zero-order coefficients uniformly $C^{0,\alpha}$, and ellipticity is uniform. All these are properties of the fixed regular charts.

For completeness the last derivative can be obtained from the Weitzenböck equation. In coordinates its principal coefficients are uniformly $C^{1,\alpha}$, its first-order coefficients are bounded, and its zero-order coefficients consist of curvature and quadratic first metric derivatives. They are uniformly L^p

because the metrics are uniformly $W^{2,p}$. The already obtained $W^{1,p}$ and L^∞ bounds therefore put its right-hand side in L^p. Interior second-order elliptic estimates give uniform $W^{2,p}$ bounds for Ω_j. Choosing p large gives precompactness in $C^{1,\beta}$ on a smaller chart. Every limit equals ω_∞ by (5), so (4) follows. No derivative of the Ricci error was taken. □

This upgrade is specific to the harmonic representative. Applying the same conclusion directly to the auxiliary ω_j, whose small derivative is initially only an averaged estimate, would not be justified.

3. The geometric collar supplies a strong convex filling

Lemma 3 (canonical contact boundary). For each sufficiently small fixed Y_v, the restriction of Ω_j to its pulled-back collar is symplectic for large j. It has a primitive whose Liouville field points outward from $B_{v,j}$, and the induced contact structure on its boundary is contactomorphic to the canonical contact structure of the complex quotient singularity (X, x_v). If H_j is zero-free throughout $B_{v,j}$, then $(B_{v,j}, \Omega_j)$ is a strong symplectic filling with that contact boundary and the quotient-link orientation.

Proof. A uniformizing chart is a ball in $\mathbb{C}^2$ with a finite subgroup $\Gamma \subset U(2)$ acting freely away from the origin. The limiting smooth invariant Kähler metric has an invariant strictly plurisubharmonic local potential φ. After subtracting an invariant pluriharmonic term, φ has a positive quadratic leading part. Small regular level sets provide links Y transverse to the radial direction. The convention for d^c is chosen so that $\omega_\infty = dd^c\varphi$. The primitive $\alpha_\infty = d^c\varphi$ gives a positive contact form on Y with an outward-pointing Liouville field. Its contact structure is the canonical quotient contact structure: interpolation of the strictly plurisubharmonic potentials on a sufficiently small chart and Gray stability identify it with the link of the standard quadratic potential.

Fix a compact collar $C = Y \times [-b, b]$ in the regular locus. Since Y is a spherical space form, $H^2(C;\mathbb{R}) = H^2(Y;\mathbb{R}) = 0$. On a slightly smaller collar closed two-forms admit a continuous linear right inverse to d. One concrete construction uses a fixed Hodge primitive on a slice Y and integration of the normal contraction along the collar coordinate. Consequently the closed difference $\Omega_j - \omega_\infty$ has primitives $\gamma_j \to 0$ in C^1, by Lemma 2. Put

$$\alpha_j = \alpha_\infty + \gamma_j, \qquad d\alpha_j = \Omega_j. \tag{7}$$

The contact and outward-transversality conditions are open; (4) and (7) preserve them for large j. Gray stability identifies the boundary contact structures. The collar is fixed first and j tends to infinity afterwards; no estimate uniform as its radius tends to zero is asserted.

Finally $d\Omega_j = 0$, and self-duality gives

$$\Omega_j \wedge \Omega_j = |\Omega_j|^2 \, d\mu_{g_j}. \tag{8}$$

If the form has no zero on $B_{v,j}$, (8) makes it symplectic on the entire filling in its existing orientation. Its strong convex boundary is supplied by (7). In particular the orientation is the singularity-link orientation, not its reverse. □

4. Classical filling classification determines the whole manifold

Lemma 4 (the two filling alternatives). Let B be a strong symplectic filling of the canonical contact link of an isolated complex quotient surface singularity.

(a) If the singularity is A_1 and $b_2(B;\mathbb{Q}) = 1$, then B is oriented diffeomorphic to the (-2)-disk bundle over S^2, hence to the A_1 Milnor fiber.

(b) If B is a rational homology ball, the singularity is Wahl and B is oriented diffeomorphic to its Q-Gorenstein Milnor rational ball.

Proof. The link in (a) is canonical contact $L(2,1)$. McDuff's filling theorem, stated in the precise form used here in Lisca [L08, Introduction], says that every such filling is an oriented smooth blowup of the (-2)-disk bundle. Each blowup increases b_2 by one, so the assumed value excludes every blowup. The quadratic A_1 Milnor fiber is the cotangent disk bundle of S^2, with this oriented smooth type.

For (b), Etnyre–Ozbagci–Tosun [EOT25, Theorem 1.18] applies to a spherical link **with the quotient-singularity orientation**. It forces the oriented link to be a lens space

$$L(n^2, na-1), \qquad 0 < a < n, \quad \gcd(n,a) = 1. \tag{9}$$

The trivial spherical group is not at issue here, since the point is a singularity. The fundamental group of the link is the quotient group, so that group is cyclic. Its faithful complex representation is diagonal and, because it acts freely on the three-sphere, the complex singularity is $1/P(1,Q)$. Oriented lens-space classification gives $P = n^2$ and

$$Q \equiv na-1 \quad \text{or} \quad (na-1)^{-1} \pmod{n^2}.$$

The inverse is $n(n-a)-1$ modulo n^2; either representation is a Wahl singularity, differing at most by interchanging its coordinates.

For the standard contact structure on (9), the symplectic rational-ball filling is unique up to symplectic deformation, in particular up to oriented diffeomorphism: [GS22, Proposition A.2], derived from Lisca's classification. Its known representative is the Wahl Milnor rational ball. The latter is a Stein filling and the $d = 1$ case of the Q-Gorenstein deformation model in [S11, Proposition 2.3]. This identifies the entire B, not one selected ALE subdomain. □

5. Arbitrary actual boundary markings extend

Classification up to unmarked diffeomorphism alone would leave the old gluing issue unresolved. In the preceding two alternatives that issue can be settled explicitly.

Lemma 5 (extension of the boundary correction). Every orientation-preserving self-diffeomorphism of the boundary of an A_1 Milnor fiber or a Wahl Milnor rational ball extends to an orientation-preserving diffeomorphism of that filling.

Proof. For A_1, the boundary is $\mathbb{RP}^3$. Bonahon [B83, Theorem 3(e)] implies that every orientation-preserving self-diffeomorphism is isotopic to the identity: the other component reverses orientation. Extend such an isotopy through a collar, taking the identity farther inside.

For a Wahl point write $P = n^2$, $Q = na-1$. If $n > 2$,

$$Q^2 \equiv 1 - 2na \pmod{n^2}, \qquad Q^2 \not\equiv \pm 1 \pmod{n^2}. \tag{10}$$

Indeed the plus case would imply $n \mid 2a$, hence $n \mid 2$, and the minus case already implies $n \mid 2$ after reducing modulo n. Bonahon's Theorem 3(a) gives two mapping classes, represented by the

identity and simultaneous complex conjugation. For $n = 2$, $a = 1$, $(P, Q) = (4, 1)$, and Theorem 3(c) gives exactly the same conclusion. Both maps preserve three-dimensional orientation.

Use the standard real-parameter Wahl Milnor model

$$F_{n,a,t} = \{(x, y, z) \in \mathbb{C}^3 : xy = z^n + t\}/\mu_n, \qquad t > 0, \tag{11}$$

cut off by a sufficiently small invariant Milnor ball. The action is

$$\xi \cdot (x, y, z) = (\xi x, \xi^{-1} y, \xi^a z), \qquad \xi^n = 1. \tag{12}$$

It is free on the smooth fiber. Complex conjugation preserves (11), normalizes (12) by $\xi \mapsto \xi^{-1}$, and preserves the cutoff. It thus extends over the quotient fiber. It preserves the real orientation because the fiber has complex dimension two. The usual equivariant Milnor collar identification, along real parameters, identifies its boundary action with conjugation on the quotient link. Such a collar identification can be made equivariant by averaging its horizontal vector field over conjugation before integrating along the real parameter. In particular it acts by -1 on $H_1(L(n^2, na - 1); \mathbb{Z})$, and represents the nontrivial class in Bonahon's list. Every boundary diffeomorphism is isotopic to one of these two extendible maps; its remaining isotopy extends through a collar. □

Completion of Theorem 1. Lemmas 2 and 3 give canonical strong symplectic fillings. Lemma 4 supplies oriented diffeomorphisms from those whole fillings to the appropriate Milnor fibers. Compare the boundary map of any such diffeomorphism with the actual exterior marking and the Milnor marking. Their difference is an orientation-preserving self-diffeomorphism of the link. Extend that correction by Lemma 5 and compose. Collar adjustment makes the maps agree on neighborhoods of the boundaries, so they glue smoothly to the given exterior identification. This is exactly the relative matching required in the previous Theorem C. □

6. A realization theorem and a zero-set obstruction

Corollary 6 (negative KE realization with forced local matching). Assume (1) and the two hypotheses of Theorem 1. If the limiting surface X has a projective Q-Gorenstein smoothing, then every sufficiently large M_j in this subsequence admits a negative Kähler–Einstein metric, on its original smooth structure.

A separately checkable sufficient condition for this global smoothing is

$$H^2(Z, T_Z(-\log E)) = 0, \tag{13}$$

where $Z \to X$ is a resolution with reduced exceptional divisor E, as in Lee–Park's global smoothing criterion [LP07, Theorem 2.1]. The local class-T condition in that criterion holds by Theorem 1.

Proof. Fix a small regular parameter in a global smoothing. Its exterior is diffeomorphic to K by Ehresmann trivialization, and the local fibers are the A_1 or Wahl fibers in Theorem 1. The boundary extension argument makes its gluing match the actual gluing in M_j. Thus M_j is oriented diffeomorphic to the smooth algebraic fiber. The negative KE orbifold metric makes K_X ample. A Cartier multiple of the relative canonical class remains ample on nearby smooth fibers; the Aubin–Yau theorem supplies their negative KE metrics. Pullback gives the metric on M_j. This is Theorem C of the preceding note with its local matching premise now proved, rather than assumed. □

Corollary 7 (topology forces a zero of the actual harmonic form). Under (1), if a whole filling $B_{v,j}$ is a rational homology ball but the limiting singularity is not Wahl, then, for large j, H_j has a zero in the interior of that filling. Its boundary collar is nevertheless zero-free.

Proof. Zero-freeness would give a strong symplectic rational-ball filling by Lemma 3, contradicting Lemma 4(b). Lemma 2 excludes zeros on the fixed collar. □

This is a genuine restriction on how the actual class can pass through singular topology. It is not an Einstein nonexistence theorem: a zero inside a shrinking core can be consistent with all the averaged estimates.

7. Nonempty local models and the precise gain

The additional geometric condition occurs in actual smooth metrics, not just formal cohomology. On an A_1 Milnor fiber the Eguchi–Hanson Kähler form is closed, self-dual and nowhere zero. It supplies the first filling alternative. On each Wahl Milnor fiber, the Ricci-flat Kähler metrics supplied by Şuvaina [S11, Theorem A] have the same properties and supply the rational-ball alternative. Their Kähler forms have canonical convex boundaries on sufficiently large truncated ends. These are classical local calibration examples, not new global Yau26 constructions.

For a global negative KE input with its compatible canonical Spin^c class, $\Omega = -2\pi H$ is exactly the Kähler form and has no zeros; the two defects in (1) vanish. Accordingly, whenever a classical negative KE degeneration has A_1 Milnor neighborhoods or Wahl Milnor rational balls, this criterion passes its exact-geometry test. We do not infer a bounded-diameter degenerating sequence merely from the existence of a local ALE model.

The topological restrictions in Theorem 1 cannot simply be dropped from the **symplectic classification step**. Canonical contact $L(4,1)$ has both the (-4)-disk-bundle filling and the Wahl rational-ball filling, as recorded by McDuff/Lisca [L08, Introduction]. Both are symplectic but have different second Betti numbers. Contact geometry alone therefore does not select the Milnor rational ball. This is not being offered as a counterexample satisfying the global near-Einstein hypotheses (1).

The present result goes beyond classifying a terminal bubble in three specific ways: it controls the whole actual filling, derives the relevant contact boundary from the actual gauge-theoretic form, and removes every boundary-marking ambiguity in the two stated cases. Within these topological alternatives, it leaves two additional questions: whether the actual harmonic form is zero-free in the core, and whether the local smoothings globalize. Neither is replaced by the assertion that an ALE limit exists.

8. Sources and dependency record

- Project analytic input: actual-form composition, Theorem A, especially actual harmonic alignment (5); and orbifold smoothing bridge, Theorems A and C. Their complete hypotheses are retained above.
- [L08] P. Lisca, *On symplectic fillings of lens spaces*, Transactions of the American Mathematical Society 360 (2008), 765–799. Author manuscript. Introduction: the exact McDuff filling statement for $L(p,1)$. Theorem 1.1 and Corollary 1.2: lens-space filling classification.
- [EOT25] J. Etnyre, B. Ozbagci and B. Tosun, *Symplectic rational homology ball fillings of Seifert fibered spaces*, Advances in Mathematics (2025), article 110572. Author manuscript, Theorem 1.18 and Remark 1.21; published DOI. The orientation restriction is essential and is supplied by Lemma 3.
- [GS22] M. Golla and L. Starkston, *The symplectic isotopy problem for rational cuspidal curves*, Compositio Mathematica 158 (2022), 1595–1682, Appendix A, Proposition A.2. Published

paper. The cited uniqueness is of the whole rational-ball symplectic filling.

- [B83] F. Bonahon, *Difféotopies des espaces lenticulaires*, Topology 22 (1983), 305–314, Theorem 3(a), (c), (e). Original paper. Lemma 5 adds the explicit extension over the relevant Milnor model.
- [S11] I. Şuvaina, *ALE Ricci-flat Kähler metrics and deformations of quotient surface singularities*, Theorem A and Proposition 2.3. Author manuscript. The local algebraic model used in (11)–(12) is the $d = 1$ case.
- [LP07] Y. Lee and J. Park, *A simply connected surface of general type with* $p_g = 0$ *and* $K^2 = 2$, Inventiones Mathematicae 170 (2007), 483–505, Theorem 2.1. Author manuscript. This supplies (13), not the automatic contact or smooth matching.

Status: manuscript theorem proved by the stated analytic estimates and classical classification inputs. This note makes no worldwide priority claim, does not assume exact Einstein equations for the input metrics, and does not claim that (1) alone implies smooth realization.